



Spatial Optimization of Municipal Solid Waste Collection Infrastructure Using the Christaller Hexagonal Model: Evidence from Los Olivos District, Lima, Peru

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Abstract: Efficient spatial allocation of municipal solid waste (MSW) collection infrastructure is essential for improving urban environmental management, service accessibility, and territorial equity in high-density metropolitan areas. In this study, the territorial and operational performance of MSW collection point optimization was assessed through the application of a Christaller-based hexagonal model in Los Olivos District, Lima, Peru. A quantitative, applied research approach was adopted under a comparative cross-sectional and non-experimental design, in which the existing collection configuration was systematically compared with an optimized spatial scenario. The baseline system consisted of 23 temporary accumulation/storage points (K3) distributed under operationally functional but territorially heterogeneous conditions, whereas the optimized configuration comprised 59 proposed K3 points spatially redistributed to enhance accessibility and service uniformity. The optimized network was generated using a regular hexagonal tessellation with a circumradius of $R = 300$ m per K3 point, corresponding to a pedestrian service radius/apothem of ~ 260 m. Each hexagonal service unit covers approximately 0.234 km^2 , which was subdivided into 16 analytical zones. Spatial modelling and geostatistical processing were conducted using official cartographic datasets, population density layers, collection route information, Geographic Information Systems (GIS), Euclidean distance matrices, and geometric optimization procedures. Performance was evaluated using key indicators, including K2–K3 distance, K3–main road distance, estimated collection time, average MSW load per K3 point, population coverage, and territorial coverage. Statistical consistency was assessed using paired Student's *t*-tests and Wilcoxon signed-rank tests with a significance threshold of 1%. Under the optimized scenario, the average K2–K3 distance was reduced from 413.78 m to 224.38 m, corresponding to a 45.76% reduction. The average MSW load per K3 point decreased by 61.03%, estimated sector-level collection time decreased from 11.50 min to 6.79 min, and territorial coverage increased from approximately 32% to 77%. Population coverage also increased from 33.17% to 78.88%. These findings suggest that the Christaller-based hexagonal model provides a scalable and reproducible spatial planning framework for optimizing MSW collection infrastructure in densely populated urban districts.

Keywords: Urban solid waste management; Christaller hexagonal model; Spatial optimization; Urban planning, Waste collection systems; Geospatial analysis; Territorial equity; Operational efficiency

1 Introduction

The Christaller hexagon, derived from Christaller's central place theory [1], constitutes a geometric spatial model that optimizes territorial coverage without gaps or overlaps, maximizing accessibility and distributive equity. Later studies further extended central place theory to market-center distribution and spatial economic organization [2, 3]. The optimal location of municipal solid waste (MSW) temporary accumulation/storage points involves the strategic identification of spaces for temporary waste storage, minimizing distances, logistical costs, and critical waste accumulations that compromise public health [4, 5]. The relationship between Christaller's hexagonal model and MSW temporary accumulation/storage points lies in the fact that the hexagonal pattern functions as a structural framework to spatially rationalize the location of collection points, transforming an administrative task into a

geometrically efficient process [6, 7]. The hexagonal model supports uniform coverage of the territory, surpassing circular or quadrangular models by eliminating overlaps and exclusions [8]. This theory has proven effective in the location of urban infrastructure in Europe, Asia, and Latin America [9]. The location of MSW temporary accumulation/storage points is particularly important in large metropolitan areas such as Lima. In Peru, more than 20,000 tons of municipal solid waste are generated daily, approximately 70% of which comes from households. In addition, 631 critical garbage accumulation points were identified in Lima Metropolitana and Callao as of September 2018, including 595 in Lima Metropolitana and 36 in Callao [10]. Its relevance lies in the reduction of health risks, environmental mitigation, and regulatory compliance with Peru's Legislative Decree No. 1278 [11], which approves the Law on Integrated Solid Waste Management, and the Sustainable Development Goals (SDGs), particularly Target 11.6 on municipal waste management and Target 12.5 on prevention and recycling [12]. Integrating Christaller's hexagonal model with the siting of MSW temporary accumulation/storage points therefore represents a methodological innovation: applying classic spatial planning to contemporary environmental problems, generating systemic, replicable and geospatially grounded solutions.

In Los Olivos, the irregular distribution of waste collection points has contributed to persistent critical waste accumulations exceeding 1.5 m^3 , with serious health and environmental implications. The OEFA (Organismo de Evaluación y Fiscalización Ambiental)'s *Reporta Residuos* platform ranks Los Olivos among the three districts with the highest number of unattended waste alerts in Metropolitan Lima, alongside Villa El Salvador and Puente Piedra [13]. This indicates a clear methodological gap: although Peruvian studies and institutional reports have addressed municipal solid waste management, the application of a Christaller-inspired hexagonal framework to the siting of MSW temporary accumulation/storage points remains underexplored. This gap limits the development of proactive, equitable, and spatially systematic planning approaches in densely populated urban districts.

Central place theory [14] has been a cornerstone of urban planning since its inception, and has been widely validated in multiple contexts. Berry [2] applied it in the United States for the distribution of commercial services, while Krugman [3] integrated it into geographic economics models. In Latin America [15], it was used to locate health centers in Colombia, achieving a 32% reduction in access distances. The model was adapted in Mexico for urban parks, and in Portugal for recycling points, achieving a population coverage of 94% within 500 m [16].

In Asia, Christaller hexagonal model, integrated with Geographic Information Systems (GIS), has been used to delimit the hierarchical levels of market zones in Seoul, as well as to optimize the location of waste collection points in Singapore, demonstrating its usefulness in urban management [17]. In Peru, studies on solid waste generation and management have been predominantly reactive, based on diagnoses of critical points or administrative criteria, without incorporating rigorous spatial modeling that allows for proactive and systemic planning of the collection infrastructure. The World Bank warns that global solid waste generation will increase by approximately 70% by 2050, disproportionately affecting vulnerable populations with limited access to sanitation and adequate urban services [18]. Furthermore, recent research on spatial optimization using Voronoi tessellations and geometric models applied to solid waste demonstrates that territorial reorganization can reduce operational dispersion by between 7% and 55%, improving coverage, logistical efficiency, and reducing costs in the collection and transport of urban waste [19]. Despite these warnings, in the Peruvian context there is no proposal that links the Christaller hexagon with the location of waste collection points. As can be seen, the hexagonal model of Christaller has not been applied to optimize the location of solid waste collection points and ensure sustainable urban management within the framework of the United Nations SDGs.

To the authors' knowledge, this research represents one of the first applications in Peru of the Christaller hexagonal model as a geometric approach to determine the optimal spatial location of the temporary points of accumulation and collection of MSW. Unlike qualitative approaches or those based on subjective criteria, this model offers a quantitative, replicable methodology based on spatial geometry, with transformative potential for urban environmental management in contexts of unplanned growth.

In this sense, the present research aims to evaluate the performance of the Christaller hexagonal model as a theoretical-spatial tool for optimizing the urban solid waste collection system by comparing the current scheme for locating collection points with a spatially optimized scenario. The study analyzes the system's operational efficiency based on quantifiable indicators of average distance, collection time, and load per point, as well as its impact on the service's territorial and population coverage. Furthermore, it evaluates the model's capacity to reduce the urban impacts associated with waste accumulation through a more balanced spatial redistribution of the operational load. Finally, the structural consistency of the territorial and functional indicators between both scenarios is examined, in order to determine whether the improvements observed respond to a systematic and technically sound spatial organization, and not to random variations of the system.

2 Materials and Methods

This article presents an applied research study with a quantitative approach and a non-experimental, cross-sectional, descriptive and correlational-comparative design. The main objective was to simulate the Christaller hexagonal model

to optimize the location of MSW temporary accumulation/storage points (K3) in the Los Olivos district of Lima, Peru.

The district of Los Olivos has an area of 18.25 km² and a population of 366,751 inhabitants based on the DIRIS Lima Norte (Dirección de Redes Integradas de Salud Lima Norte) [20], resulting in a population density of approximately 20,096 inhabitants/km². This population generates 350 tons of MSW per day. Los Olivos has a maximum length of 10 km and an average width of 1.83 km (Figure 1). As one of the 43 districts of Metropolitan Lima, Los Olivos is characterized by predominantly flat urban terrain and a hierarchical road network, which facilitates the application of spatial optimization models.



Figure 1. Location of Los Olivos district in Lima Norte, Lima, Peru

The district was divided into 16 urban sectors for comparative quantitative analysis. For each sector, area and population were compiled as the basic spatial and demographic inputs (Table 1). The total area calculated using GIS for the 16 sectors is 18.225 km². These data were then used to calculate population density, estimated solid waste generation, waste density, and territorial criticality levels. In the subsequent spatial analysis, average distances among the hierarchical MSW management levels K1, K2, K3, K4, and K5 were also calculated to evaluate accessibility and operational efficiency. In this study, K1 represents solid waste generation points, K2 represents primary collection points or containers, K3 represents temporary accumulation/storage points, K4 represents composting points, and K5 represents transfer points. These indicators supported the establishment of numerical criticality thresholds based on

population concentration, effective solid waste load, accessibility, and operational performance.

Table 1. Urban sectors of Los Olivos

No.	Urban Sector	Area (km ²)	Population
1	Urb. Prolima	3.121	45,000
2	Urb. San Elias	2.363	43,850
3	Urb. Villa del Norte	0.697	29,135
4	Urb. Parque Naranjal	1.196	39,072
5	Urb. Villasol	0.894	22,029
6	AA.HH. Laura Caller Ibérico	0.764	18,640
7	Conj. Hab. Carlos Cueto Fernandini	1.292	16,741
8	Urb. 12 de Agosto (Mariscal R. Castilla)	0.798	18,217
9	Urb. El Trébol	1.375	26,280
10	Urb. Previ Naranjal	0.595	11,470
11	Urb. Villa Los Ángeles–Sol de Oro	0.824	22,814
12	Urb. Covida	0.995	20,327
13	Urb. Santa Rosa de Lima (Virgen de La Puerta)	0.712	13,812
14	AA.HH. Villa Mercedes (Urb. 28 de Julio)	0.498	10,102
15	Urb. Parque Industrial Naranjal	1.651	19,237
16	Urb. Mercurio	0.450	10,025
	Total	18.225	366,751

Note: Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex. Sector areas were calculated from Geographic Information Systems (GIS)-based sector boundaries and rounded to three decimal places.

In ArcGIS 10.8, the population table was joined to the sector layer using the common sector ID. Sector areas were calculated in square kilometers using the Calculate Geometry tool. Population density was then calculated by dividing the sector population by its area, while waste density was calculated by dividing the estimated waste generation by the sector area. The resulting spatial layer was used to map the distribution of municipal solid waste generation in Los Olivos (Figure 2).

The location of the initial K3 points was determined based on the urban centroid, which was identified as the core area with high MSW generation and favorable road accessibility. From this central node, a hierarchical hexagonal grid derived from the Christaller's central place theory was superimposed on the district territory, allowing the other points to be spatially structured according to criteria of coverage, equidistance, and functional efficiency.

The hexagonal spatial unit was defined with a side length L and circumradius R of 300 m, resulting in an area of approximately 0.2338 km². In a regular hexagon, the pedestrian service radius corresponds to the apothem ($L \times \sqrt{3}/2 \approx 259.8$ m). This value is broadly consistent with the 100–300 m range reported in GIS-based MSW collection studies for recyclable drop-off or temporary collection points, although optimal collection-point distances are site-specific and may vary according to location, urban form, building type, and waste composition [20]. The selected dimensions were determined according to the district's effective urban area of 13.688 km² and operational coverage criteria. Given the optimal operating side length of $L = 300$ m, which is equivalent to the circumradius and corresponds to a pedestrian service radius of approximately 260 m, the area of a regular hexagon was calculated as follows:

$$A_{\text{hex}} = \frac{3\sqrt{3}}{2} \times 300^2 \approx 0.2338 \text{ km}^2$$

The theoretical number of required K3 temporary accumulation/storage points was calculated as:

$$N_{K3} = \frac{A_{\text{eff}}}{A_{\text{hex}}} = \frac{13.688 \text{ km}^2}{0.2338 \text{ km}^2} \approx 58.55$$

where, $A_{\text{eff}} = 13.688 \text{ km}^2$ is the consolidated urban area of Los Olivos district, excluding non-urbanizable zones, and A_{hex} is the area of one regular hexagonal unit.

Therefore, the theoretical value of N_{K3} was rounded up to 59 representing the minimum number of K3 points required to ensure continuous territorial coverage without spatial gaps and with improved operational efficiency (Figure 3).

Based on the geometry of Christaller's central place theory, a hierarchical waste management system was defined. Specifically, 81,541 K1 points, 4,584 K2 points, and 59 K3 points were structured within the district. The K3 points were arranged using a regular hexagonal grid with a circumradius of $R = 300$ m. From this level, seven K4 composting plants ($R \approx 520$ m) and two K5 transfer stations ($R \approx 900$ m) were estimated, as shown in Table 2.

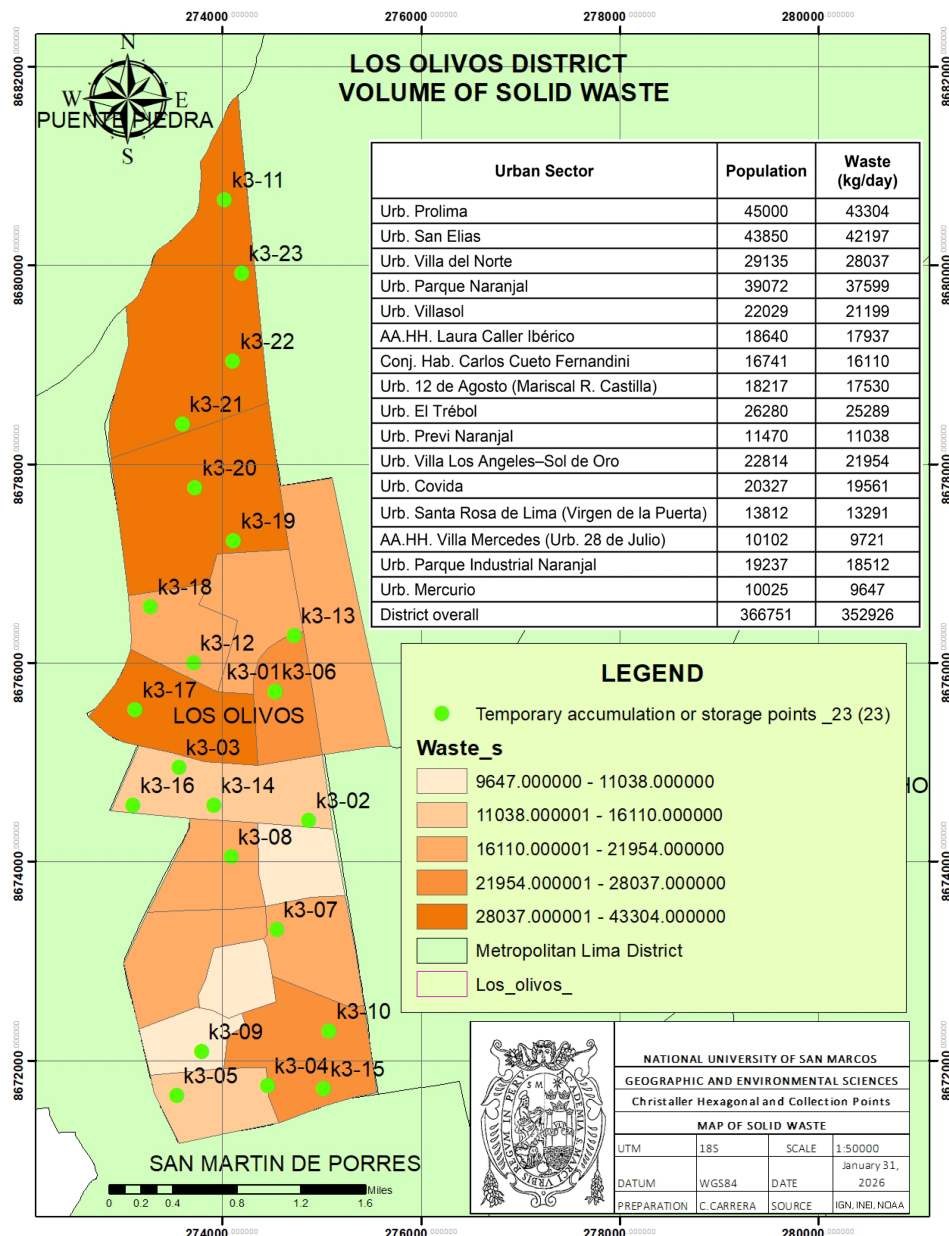


Figure 2. Map of the Los Olivos district showing the volume of solid waste by zone, with temporary accumulation/storage points (K3) and daily tons of garbage generated

Table 2. Hierarchical structure of the solid waste management system based on Christaller's central place theory for the district of Los Olivos, Lima

Level	Function	Number of Points	Circumradius <i>R</i> (m)	Feeds into / Supplies to
K1	Solid waste generation points	81,541	—	Feeds K2
K2	Primary collection points or containers where citizens deposit waste	4,584	100–150*	Feeds K3
K3	Temporary accumulation/storage points where collection trucks load waste	59	300	Feeds K4
K4	Composting points	7	520	Feeds K5
K5	Transfer points	2	900	Sends to final disposal

Note: “—” indicates not applicable. * The 100–150 m range for K2 is route-dependent and provided for reference only.

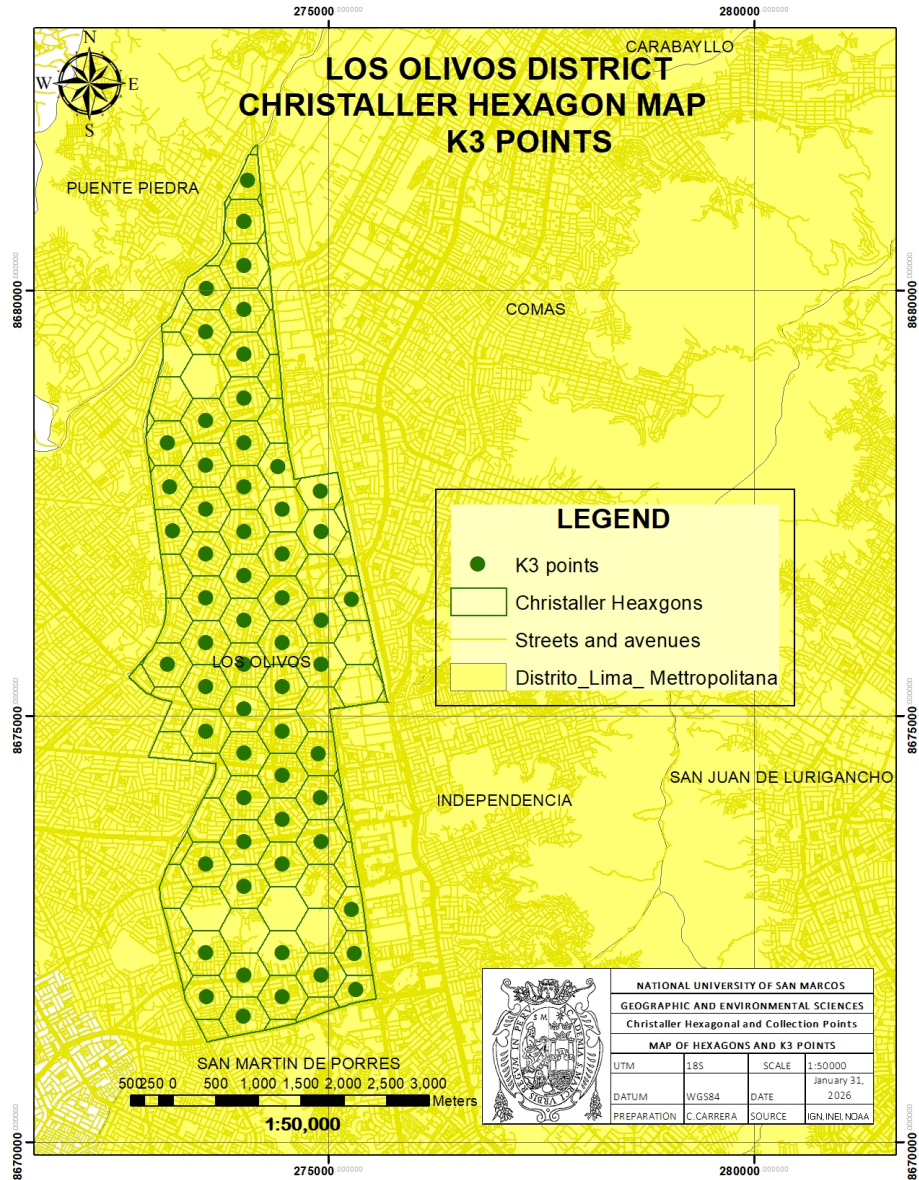


Figure 3. Christaller hexagon distribution map

Note: K3 = temporary accumulation/storage points. The 59 K3 points were generated in ArcGIS 10.8 using a hexagonal grid with circumradius $R = 300$ m. The grid was intersected with the consolidated urban area and adjusted according to accessibility and territorial coverage criteria.

The hierarchical levels K1 and K2 were calculated as follows.

According to the DIRIS Lima Norte [20], Los Olivos District has a population of 366,751 inhabitants. Using an average household size of 4.5 people per dwelling, the theoretical number of K1 points was calculated as follows:

$$N_{K1} = \frac{\text{Total population}}{\text{Average residents per dwelling}}$$

With the specific values for Los Olivos:

$$N_{K1} = \frac{366,751}{4.5} \approx 81,500$$

Geospatial processing in ArcGIS 10.8, which applied the per capita generation factor by urban sector and refined the points using the centroids of the blocks weighted by population, yielded 81,541 K1 points. This value differs from the theoretical estimate of 81,500 by less than 1%, attributable to the exclusion of uninhabited or nonresidential blocks.

To determine the number of K2 points, the number of city blocks was first calculated. The number of urban city blocks in Los Olivos was estimated using the average area of the cadastral blocks (11,944 m²).

Considering that approximately 25% of the district's area corresponds to roads, parks, and public spaces, only 75% of the total district area (18.25 km²) was considered as the effective urbanized area for the city block estimation:

$$N_{\text{blocks}} = \frac{0.75 \times A_{\text{total}}}{A_{\text{block}}}$$

where, $A_{\text{total}} = 18.25 \text{ km}^2$ is the official area of Los Olivos district, and $A_{\text{block}} = 0.011944 \text{ km}^2$ is the average cadastral block area.

$$N_{\text{blocks}} = \frac{0.75 \times 18.25}{0.011944} \approx 1,146$$

Based on the operational planning criterion adopted in this study, four primary collection points/containers were assigned to each block. This assumption is consistent with the Ministry of Environment (MINAM) guidance that collection points and containers should be planned according to waste generation, container capacity, and collection-route planning. Therefore, the number of K2 points was calculated as:

$$N_{K2} = N_{\text{blocks}} \times 4.0 = 1,146 \times 4.0 = 4,584$$

Thus, 4,584 K2 points were estimated for the district.

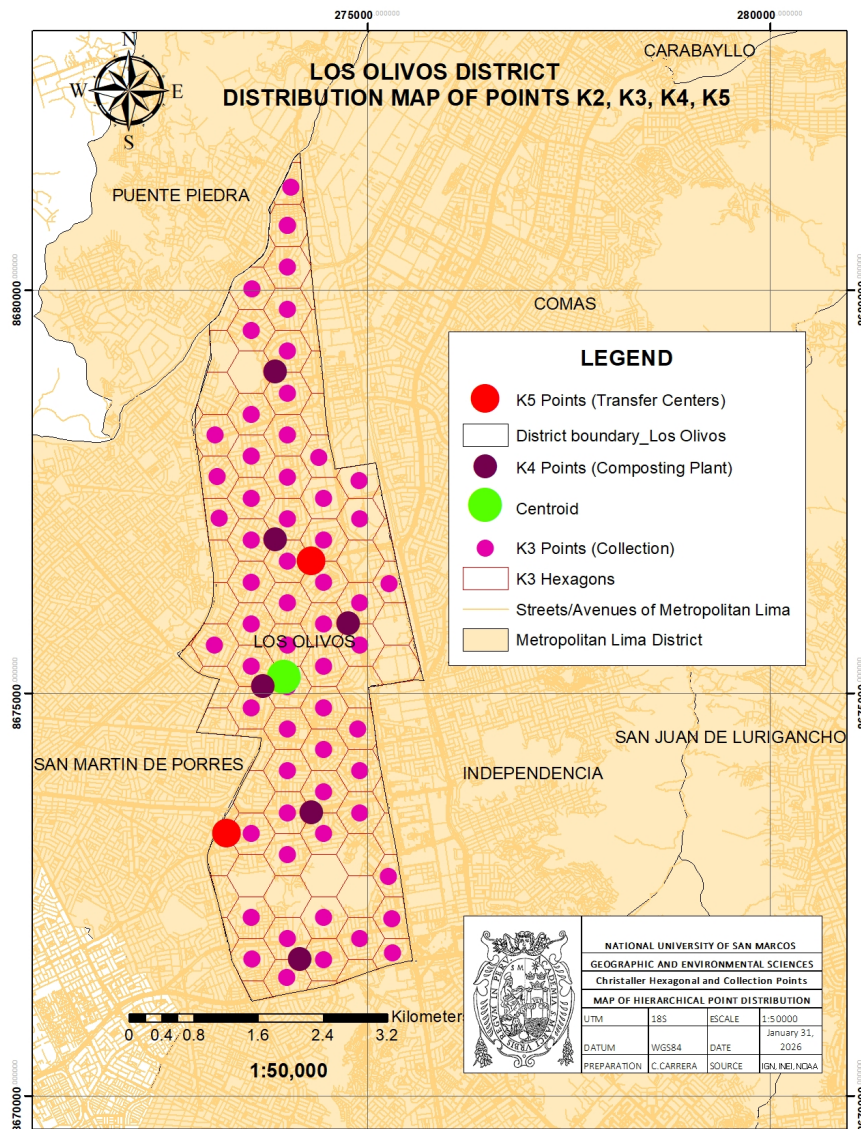


Figure 4. Distribution map of K3, K4, K5 points

Note: K3 = temporary accumulation/storage points; K4 = composting points; K5 = transfer points. The K2–K5 layers were overlaid in ArcGIS 10.8 using different symbols to show the functional and spatial organization of the proposed municipal solid waste (MSW) management system within the district.

The spatial distribution of points K2, K4, and K5 was hierarchically structured in ArcGIS 10.8. The 4,584 K2 points were located according to residential density within each K3 hexagon, ensuring homogeneous coverage. The seven K4 sites were defined by geometric aggregation of K3 sites with a radius of 520 m, while the two K5 sites were determined by gravitational optimization with a radius of 900 m. The final location of K4 and K5 was adjusted considering urban complexity, road accessibility, and land-use restrictions (Figure 4).

The performance of the proposed MSW management system was evaluated using a set of quantitative indicators calculated from explicit equations, applied comparatively to two spatial scenarios: the current scenario with 23 collection points and the optimized scenario with 59 K3 points distributed using the Christaller hexagonal grid.

The spatial analysis of Euclidean distances among K1–K5 points was conducted at two scales. The citizen scale was represented by the K1–K2 distance, while the vehicular operational scale was represented by the K2–K3, K3–K4, and K4–K5 distances. MSW collection time, territorial coverage, population coverage, and urban load per K3 point were also evaluated according to the hexagonal model.

Formal calculation equations were used for each indicator in both scenarios. Correction factors were derived from the comparison between the empirical scenario with 23 existing points and the theoretical hexagonal scenario with 59 proposed points using ArcGIS 10.8 tools.

The metrics evaluated and their calculation bases were as follows:

- Inter-level distances (K1–K2, K2–K3, K3–K4, and K4–K5), estimated using Euclidean distances in GIS and weighted averages by urban zone.
- Average distance from K3 points to main routes, calculated geometrically and integrated with the K2–K3 distance.
- Vehicle collection time, estimated based on the average MSW collection time record.
- Territorial coverage, determined using the hexagonal area of influence $A = \frac{3\sqrt{3}}{2}R^2$ and its relationship to the effective urban area of the district.
- Population coverage, obtained by spatial intersection between 300 m service buffers and population layers.
- Urban load per collection point, defined as the ratio between daily MSW generation and the number of operational points.

All equations were applied consistently in both scenarios, allowing a comparative assessment of accessibility, spatial equity, logistical efficiency, and urban load reduction.

This hierarchical structure was designed to improve waste-flow organization, reduce transport distances, and support more efficient MSW management in a high-density urban context.

Table 3. Weighted overlay scheme for generating the solid waste criticality map

Thematic Layer	Indicator Used	GIS Procedure	Decision Criterion	Weight (%)	Reclassification (1–5)
Population density	Inhabitants/km ²	Rasterization and normalization	Higher density = higher priority for accumulation points	40	1 = Very low; 2 = Low; 3 = Medium; 4 = High; 5 = Very high
Road accessibility	Euclidean distance to roads by class	Euclidean distance + Reclassification	Optimal proximity to collector roads	25	1 = >300 m; 3 = 0–50 m and 150–300 m; 5 = 50–150 m
Waste generation proxy	Urban land-use typology	Reclassification by land use	Higher urban activity = higher waste-generation potential	20	1 = Green areas; 2 = Industrial; 3 = Mixed; 4 = Commercial/medium residential; 5 = High-density residential
Distance to existing points	Service gaps from the 23 existing accumulation points	Euclidean distance + Reclassification	Farthest areas = higher priority	15	1 = <200 m; 3 = 200–300 m; 4 = 300–400 m; 5 = >400 m
Integrated result	Municipal solid waste (MSW) criticality map	Weighted overlay	Multi-criteria synthesis	100	1–5: Very low to very high criticality

Note: GIS = Geographic Information Systems. The weighted overlay integrated population density, road accessibility, waste-generation potential based on land-use typology, and distance from the 23 existing accumulation points. The resulting criticality map, classified on a 1–5 scale, was used to guide the spatial adjustment of the 59 proposed hexagonal centroids.

Regarding data sources and collection, primary and secondary data were integrated. Primary data were obtained

through field surveys using a Garmin eTrex 32x GPS receiver (accuracy ± 3 m). were obtained from official sources, including the Municipality of Los Olivos, OEFA, MINAM, and National Institute of Statistics and Informatics (INEI), covering demographic data, waste generation, urban sectors, and relevant planning information.

The geospatial analysis and processing were performed using ArcGIS 10.8 and QGIS 3.34 under the UTM Zone 18S–WGS84 coordinate reference system to ensure spatial consistency and metric accuracy in all calculations.

Four base thematic layers were created, reclassified on a 1–5 scale, and integrated using a weighted overlay procedure (Table 3). The percentages (40%, 25%, 20% and 15%) establish the weighting of each thematic layer in the weighted overlay of the GIS, generating an integrated criticality map (1–5) to prioritize strategic accumulation areas.

These layers were integrated using a weighted overlay procedure with differentiated weights, generating a spatial criticality map of solid waste ranging from 1 to 5. This map was used as an intermediate GIS layer to identify priority areas for K3 point adjustment (Figure 5). The regular hexagonal grid based on the Christaller model, with a circumradius $R = 300$ m for K3 points, was superimposed on this criticality map, establishing 59 theoretical centroids of accumulation or temporary storage points. These centroids were subsequently compared and spatially adjusted within each hexagon according to the local criticality level, while maintaining the logic of the central place model and adapting it to the actual territory.

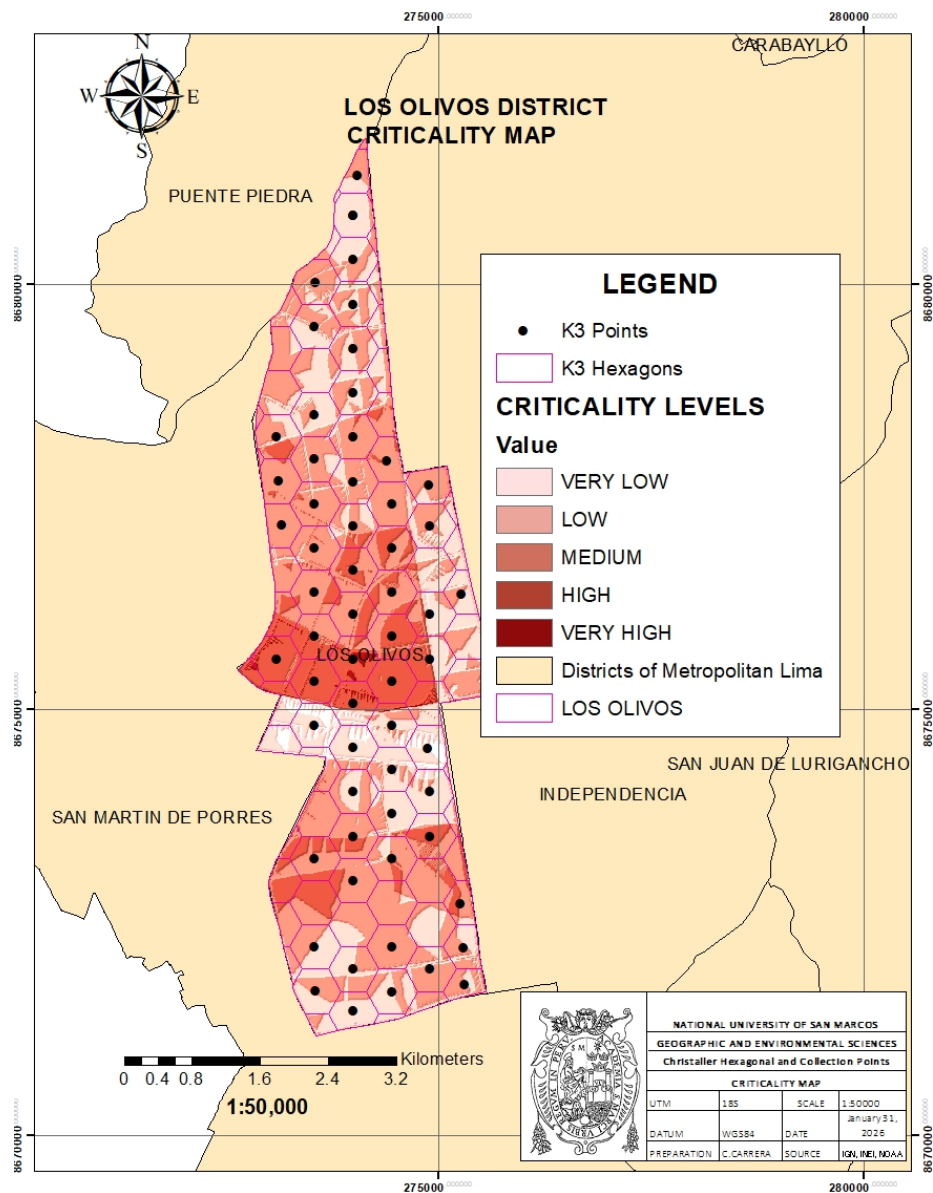


Figure 5. Spatial criticality map used for K3 point adjustment in Los Olivos

Note: K3 = temporary accumulation/storage points. The spatial criticality map was developed using weighted overlay, classifying values from 1 (low criticality) to 5 (very high), based on territorial variables and pressure from solid waste.

The statistical treatment was performed in IBM SPSS Statistics, version 26.0 (IBM Corp., Armonk, NY, USA).

Descriptive statistics and comparative tests, including Student's *t*-test and the Wilcoxon signed-rank test, were applied to compare the baseline scenario with 23 points and the optimized scenario with 59 points in terms of access distance, coverage, and spatial equity.

Finally, the operational methodological sequence was structured in the following phases: data collection and cleaning; geospatial modelling; weighted overlay analysis; construction of the Christaller-based hexagonal grid; interscale comparative analysis; and statistical comparison of the two scenarios (Figure 6). This methodology provided the basis for evaluating whether the proposed relocation and redistribution of 59 collection points could improve spatial coverage, reduce operational transport distances, and enhance citizen accessibility in the MSW management system of a high-density urban context.

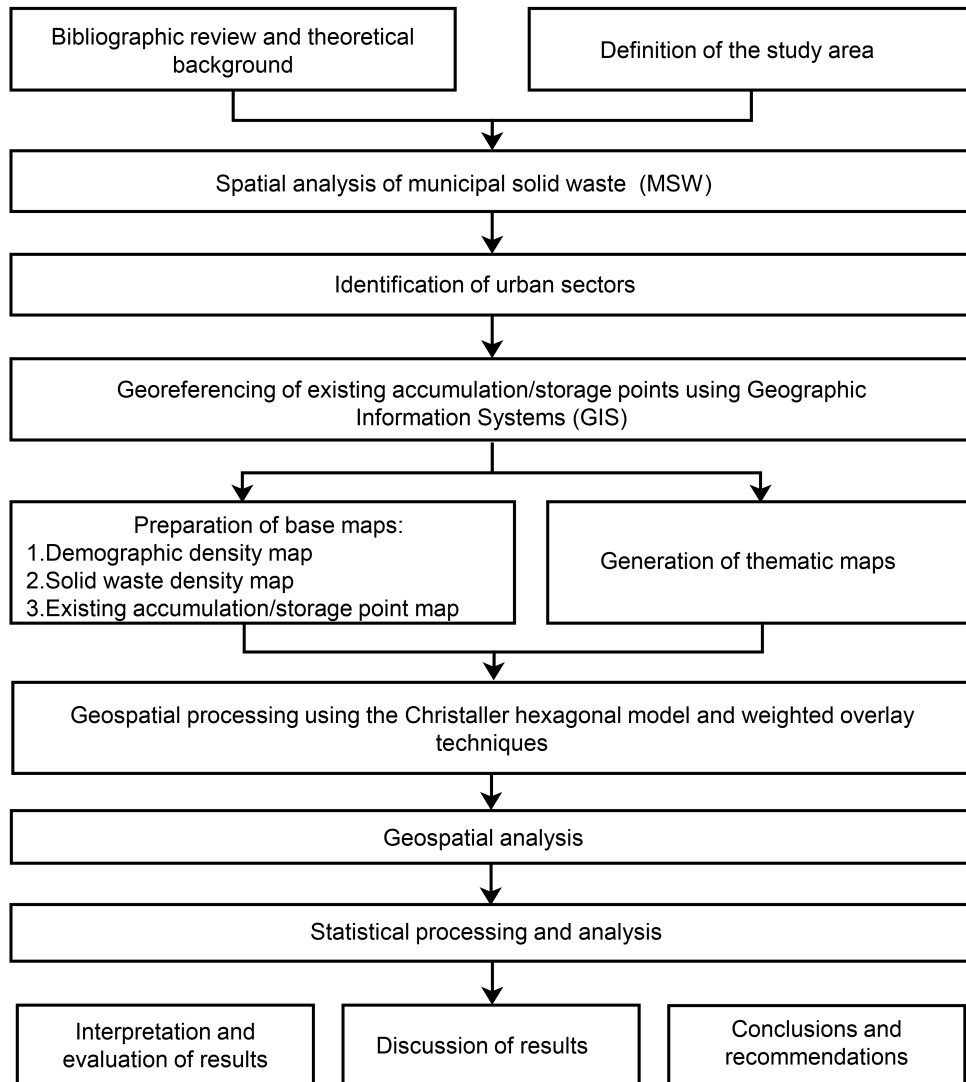


Figure 6. Methodological scheme

3 Results

3.1 Improved Operational Efficiency

3.1.1 Distance from municipal solid waste generation points to primary collection points/containers

K1 points were modelled using the centroids of 1,146 urban blocks as an approximation of waste-generation locations weighted by population. The K1–K2 Euclidean distances were calculated in ArcGIS 10.8 using the Near tool between the weighted K1 centroids and K2 points. A total of 81,541 K1 points and 4,584 K2 points were modelled across the 16 urban sectors. For each sector, the weighted distance $W_i \times Dist$ was calculated, where W_i represents waste generation and $Dist$ represents the distance to the nearest K2 point. The values were statistically normalized to allow comparison among sectors (Table 4).

Table 4. Average K1–K2 distance by urban sector

No.	Urban Sector	Distance K1–K2 (m)
1	AA.HH. Laura Caller Ibérico	61.28
2	AA.HH. Villa Mercedes (Urb. 28 de Julio)	8.78
3	Conj. Hab. Carlos Cueto Fernandini	19.21
4	Urb. Covida	9.16
5	Urb. El Trébol	40.31
6	Urb. Mercurio	49.12
7	Urb. Parque Industrial Naranjal	18.67
8	Urb. Parque Naranjal	41.03
9	Urb. Previ Naranjal	14.39
10	Urb. Prolima	20.53
11	Urb. San Elias	39.27
12	Urb. Santa Rosa de Lima (Virgen de La Puerta)	35.23
13	Urb. Villa del Norte	26.05
14	Urb. Villa Los Ángeles - Sol de Oro	50.39
15	Urb. Villasol	25.46
16	Urb. 12 de Agosto (Mariscal R. Castilla)	12.29
	Average	29.45

Note: K1 = solid waste generation points; K2 = primary collection points or containers; Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex. Values represent the average K1–K2 distance by urban sector.

Table 5. Average K2–K3 distance under the current and optimized scenarios

No.	Urban Sector	Average K2–K3 Distance (m)		Reduction (m)	Variation (%)
		Current Scenario (23 K3 Points)	Hexagonal Model (59 K3 Points)		
1	AA.HH. Laura Caller Ibérico	292.10	232.28	59.82	20.48
2	AA.HH. Villa Mercedes (Urb. 28 de Julio)	286.27	233.88	52.39	18.30
3	Conj. Hab. Carlos Cueto Fernandini	285.82	236.44	49.38	17.28
4	Urb. Covida	791.94	301.39	490.55	61.95
5	Urb. El Trébol	258.76	195.83	62.93	24.32
6	Urb. Mercurio	571.09	236.11	334.98	58.66
7	Urb. Parque Industrial Naranjal	677.95	285.45	392.50	57.91
8	Urb. Parque Naranjal	360.99	206.67	154.33	42.76
9	Urb. Previ Naranjal	416.86	185.80	231.06	55.41
10	Urb. Prolima	420.20	228.58	191.62	45.60
11	Urb. San Elias	433.32	191.40	241.92	55.84
12	Urb. Santa Rosa de Lima (Virgen de la Puerta)	287.78	192.49	95.29	33.12
13	Urb. Villa del Norte	335.09	194.59	140.50	41.93
14	Urb. Villa Los Angeles–Sol de Oro	426.49	228.88	197.62	46.33
15	Urb. Villasol	413.51	186.67	226.83	54.86
16	Urb. 12 de Agosto (Mariscal R. Castilla)	362.35	253.70	108.65	29.98
	District average	413.78	224.38	189.40	45.76

Note: K2 = primary collection points or containers; K3 = temporary accumulation/storage points; Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex. The percentage reduction represents the relative decrease in average K2–K3 distance from the current 23-point scenario to the optimized 59-point hexagonal scenario.

3.1.2 Distance from primary collection points/containers to temporary accumulation/storage points

The K2–K3 distance was calculated in ArcGIS 10.8 using the Near tool to estimate the Euclidean distance from each of the 4,584 K2 points to the nearest K3 point. Two scenarios were evaluated: the current scenario with 23 existing K3 points and the optimized scenario with 59 proposed K3 points distributed in a Christaller-type hexagonal

pattern. Summary statistics were then applied to obtain the average K2–K3 distance for each urban sector and to calculate the absolute and relative reduction between the two scenarios (Table 5).

The increase from 23 to 59 K3 points generated a substantial improvement in territorial accessibility. The average district-level K2–K3 distance was reduced from 413.78 m to 224.38 m, representing an absolute decrease of 189.4 m and a relative reduction of 45.76%.

The largest reductions were observed in Covida (61.95%), Mercurio (58.66%), Parque Industrial Naranjal (57.91%), San Elías (55.84%), Previ Naranjal (55.41%), and Villasol (54.86%). These results demonstrate that a high spatial sensitivity to nodal redistribution, and that the hexagonal configuration can improve territorial coverage and reduce spatial friction within the MSW collection system.

For comparison, a theoretical service radius was also estimated using the hexagonal model. This calculation relates the effective urban area of Los Olivos ($A_e = 13.688 \text{ km}^2$) to the number of proposed temporary accumulation/storage points ($N = 59$):

$$R = \sqrt{\frac{2A_e}{3\sqrt{3}N}} = \sqrt{\frac{2 \times 13,688,000}{3\sqrt{3} \times 59}} \approx 298.83 \text{ m}$$

This value (298.83 m) represents the theoretical circumradius of each hexagonal service area rather than the actual average K2–K3 distance. The ArcGIS Near analysis produced a lower district average of 224.38 m, resulting in a difference of 74.45 m between the theoretical service radius and the GIS-derived mean distance.

The implementation of Christaller hexagonal model suggests a potentially more efficient spatial reorganization of the waste collection system. Nodal redistribution reduces the average service distances, improved territorial coverage, and decreased spatial friction within the collection system. This reduction may contribute to shorter operational routes, lower logistical costs, and improved solid waste collection efficiency in the analyzed urban area.

3.1.3 Distance from temporary accumulation/storage points to composting points

In the functional hexagonal model applied to Los Olivos, seven K4 points (composting plants) were defined to support balanced territorial coverage and decentralized MSW management in a dense and complex urban area. The K3–K4 distance was determined in ArcGIS using the Near tool. K3 points were used as the input features, while K4 points were used as the near features. The tool calculated the Euclidean distance from each K3 point to the nearest K4 point and generated the NEAR_DIST field. Then, summary statistics were then used to obtain the average K3–K4 distance by urban sector (Table 6).

Table 6. Average distance from K3 to K4 points

No.	Urban Sector	Average K3–K4 Distance (m)		Variation (%)
		Current Scenario (23 K3 Points)	Hexagonal Model (59 K3 Points)	
1	AA.HH. Laura Caller Ibérico	791.74	698.61	–11.76
2	AA.HH. Villa Mercedes (Urb. 28 de Julio)	544.20	548.23	0.74
3	Conj. Hab. Carlos Cueto Fernandini	648.74	720.50	11.06
4	Urb. El Trébol	755.65	953.55	26.19
5	Urb. Parque Naranjal	735.48	558.63	–24.05
6	Urb. Prolima	890.98	1,093.08	22.68
7	Urb. San Elías	640.87	764.63	19.31
8	Urb. Santa Rosa de Lima (Virgen de la Puerta)	619.58	388.00	–37.38
9	Urb. Villa del Norte	319.62	401.00	25.46
10	Urb. Villa Los Angeles–Sol de Oro	309.02	715.59	131.57
11	Urb. 12 de Agosto (Mariscal R. Castilla)	575.83	601.50	4.46
	Average	621.07	676.67	8.95

Note: K3 = temporary accumulation/storage points; K4 = composting points; Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex. Values represent the average Euclidean K3–K4 distance by urban sector. Only 11 sectors are shown because five sectors had no effective nearby K3 coverage under the current 23-point scenario. The hexagonal model with 59 K3 points incorporates these previously underserved sectors through the regular grid.

Although the current 23-point scenario excludes five areas with no effective nearby K3 coverage, the hexagonal model with 59 K3 points incorporates these previously underserved areas through its regular grid. As a result, the average K3–K4 distance increases from 621.07 m to 676.67 m, corresponding to an increase of 8.95%. This increase reflects the prioritization of spatial equity and the minimization of coverage gaps over concentrated local optimization, constituting a desirable trade-off in the pursuit of an efficient and fair distribution of service across the entire territory.

Therefore, the hexagonal model provides broader territorial coverage, although with a slightly higher average K3–K4 distance.

Applying Christaller's hexagonal equation, average distance from any point within a hexagonal service area to the nearest center can be approximated as:

$$\bar{d} = 0.377\sqrt{A}$$

where, A is the area of the K4 hexagonal service area. With $A = 2.222 \text{ km}^2$, the theoretical mean distance is:

$$\bar{d} = 0.377\sqrt{2.222} \approx 0.562 \text{ km} = 562 \text{ m}$$

This value represents the theoretical mean service distance within each K4 hexagonal service area, rather than the actual ArcGIS-derived K3–K4 distance. This metric supports the planning of K3–K4 service areas by providing a theoretical reference distance for route organization, transport-cost estimation, and equitable access to decentralized MSW treatment facilities.

3.1.4 Distance from composting points to transfer stations

In the Los Olivos district, there are currently no formally defined transfer stations within the MSW management system. Therefore, the K5 points in this analysis represent proposed transfer-station locations generated within the hierarchical spatial model. In ArcGIS 10.8, the K4–K5 distance was estimated using the Near tool with the planar method, taking K4 points as the input features and K5 points as the near features. Subsequently, summary statistics were applied to calculate the average K4–K5 distance by functional service sector (Table 7).

Table 7. Average Euclidean distance from K4 to K5 points

No.	Urban Sector	Average K4–K5 Distance (m)
1	Urb. Covida	1,084.38
2	Urb. Parque Naranjal	1,674.51
3	Urb. Prolima	2,387.14
4	Urb. San Elias	520.92
5	Urb. Santa Rosa de Lima (Virgen de La Puerta)	1,804.51
6	Urb. Villa del Norte	902.25
	Average	1,395.62

Note: K4 = composting points; K5 = transfer points; Urb. = urbanization. Values represent the average Euclidean distance from K4 to K5 by functional service sector.

Although Los Olivos is divided into 16 urban sectors, only six functional K4–K5 service areas are reported in Table 7 because this level corresponds to a higher-order service hierarchy rather than the original administrative or cadastral sector division. These six areas were defined according to coverage, spatial centrality, and logistical-efficiency criteria.

Christaller hexagonal model was used to estimate the theoretical K4–K5 service distance, while ArcGIS proximity analysis was used to calculate the corresponding GIS-derived distance. The theoretical estimate was 1,386.5 m, whereas the ArcGIS-derived average was 1,395.62 m, indicating a small difference of approximately 9.12 m. This close agreement suggests that the geometric model is broadly consistent with the GIS-based measurement.

$$\bar{D}_{K4K5} = 0.537\sqrt{A_{\text{hex}, K5}}$$

Substituting $A_{\text{hex}, K5} = 6.666491 \text{ km}^2$:

$$\begin{aligned}\bar{D}_{K4K5} &= 0.537\sqrt{6.666491 \text{ km}^2} \\ \bar{D}_{K4K5} &\approx 1.3865 \text{ km}\end{aligned}$$

where, $\bar{D}_{K4K5}$ is the average theoretical distance between K4 and K5, and $A_{\text{hex}, K5}$ is the area of the K5 hexagonal service unit.

The factor 0.537 is derived from the geometry of the Christaller hexagonal model under the transport principle ($K = 4$). It represents the theoretical relationship between the average distance between hierarchical centers and the square root of the area of influence (A).

3.1.5 Distance from temporary accumulation/storage points to transfer stations

The average K3–K5 distance was obtained using a distance-matrix approach in ArcGIS 10.8. The Point Distance tool was used to generate a complete distance table between the two hierarchical levels. The resulting table was then linked to the urban sectors, and average distances were calculated to support a comparative territorial analysis between the current and optimized scenarios (Table 8).

According to the results (Table 8), the hexagonal model with 59 K3 points shows a moderate increase in the average K3–K5 distance compared with the current 23-point scenario, from 1,483.13 m to 1,568.25 m, corresponding to an increase of 5.74%. This increase reflects the incorporation of additional K3 points and broader territorial coverage within the optimized hierarchy.

Besides, several sectors show distance reductions, including AA.HH. Laura Caller Ibérico (−26.90%), Urb. Parque Naranjal (−15.35%), and Urb. 12 de Agosto (−5.72%). These localized reductions suggest that the hexagonal model may improve accessibility in selected peripheral or underserved areas, while the district-level average increases slightly due to broader spatial inclusion. These results indicate a possible trade-off between operational distance minimization and spatial equity in the provision of integrated MSW management services.

Table 8. Average Euclidean distance from K3 to K5 points

No.	Urban Sector	Average K3–K5 Distance (m)		Variation (%)
		Current Scenario (23 K3 Points)	Hexagonal Model (59 K3 Points)	
1	AA.HH. Laura Caller Ibérico	955.64	698.61	−26.90
2	AA.HH. Villa Mercedes (Urb. 28 de Julio)	1,277.59	1,294.07	1.29
3	Conj. Hab. Carlos Cueto Fernandini	1,625.22	1,750.87	7.73
4	Urb. El Trébol	2,105.61	2,133.70	1.33
5	Urb. Parque Naranjal	1,619.75	1,371.08	−15.35
6	Urb. Prolima	2,900.17	3,171.50	9.35
7	Urb. San Elias	937.19	1,157.92	23.56
8	Urb. Santa Rosa de Lima (Virgen de la Puerta)	1,631.11	1,832.67	12.35
9	Urb. Villa del Norte	820.27	1,103.22	34.49
10	Urb. Villa Los Angeles–Sol de Oro	1,291.73	1,652.77	27.95
11	Urb. 12 de Agosto (Mariscal R. Castilla)	1,150.13	1,084.38	−5.72
	Average	1,483.13	1,568.25	5.74

Note: K3 = temporary accumulation/storage points; K5 = transfer points; Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex. Values represent the average Euclidean K3–K5 distance by urban sector. Positive values in the variation column indicate an increase in distance, whereas negative values indicate a reduction. Only sectors with comparable K3–K5 distance values under both scenarios are reported.

3.1.6 Average distance from temporary accumulation/storage points to main roads

The following equation, derived from Christaller hexagonal model, was used to provide a preliminary theoretical estimate of the average distance between K3 points and the main road network. This estimate incorporates the effective service area and the total length of the main road network, thereby adapting the ideal hexagonal structure to the urban road configuration of Los Olivos.

$$\bar{D}_{K3-MRL} = k \sqrt{\frac{A_{\text{eff}}}{L_{\text{MRL}}}}$$

where, $\bar{D}_{K3-MRL}$ is the estimated average distance from K3 points to the nearest main road, A_{eff} is the effective service area, L_{MRL} is the total length of the main road network, and k is an empirical configuration coefficient for the hexagonal system. In this study, $k = 0.28$ was adopted.

$$\bar{D}_{K3-MRL} = 0.28 \sqrt{\frac{13.688 \text{ km}^2}{54.6 \text{ km}}} \approx 0.14019 \text{ km} = 140.19 \text{ m}$$

The estimated value of 140 m reflects the high main-road density within the effective urban area of Los Olivos. This value should be interpreted as a theoretical proximity estimate rather than as the actual GIS-derived distance. In ArcGIS 10.8, the Near tool was then applied using K3 points as input features and the main road network as the linear reference layer under a UTM projection. Sector-level averages were calculated using summary statistics of the NEAR_DIST field (Table 9).

On average, the K3–main road distance decreased from 105.94 m in the current 23-point scenario to 100.55 m in the hexagonal 59-point scenario, corresponding to a reduction of 5.09%. However, several sectors show large

percentage increases because their baseline distances in the current scenario were extremely low, such as Urb. Villa del Norte and Urb. 12 de Agosto.

It should also be noted that five urban sectors lacked effective nearby K3 points in the current 23-K3 scenario. Therefore, Table 9 reports the sectors with comparable distance values under both scenarios, while the optimized 59-K3 hexagonal scenario is designed to provide coverage across all 16 urban sectors. Although distances increase in some cases, the hexagonal configuration improves overall spatial continuity and may enhance territorial equity and functional accessibility within the MSW collection system.

Table 9. Average Euclidean distance from K3 points to main roads

No.	Urban Sector	Average K3–Main Roads Distance (m)		Variation (m)	Change (%)
		Current Scenario (23 K3 Points)	Hexagonal Model (59 K3 Points)		
1	AA.HH. Laura Caller Ibérico	150.62	62.91	−87.71	−58.23
2	AA.HH. Villa Mercedes (Urb. 28 de Julio)	187.36	179.02	−8.34	−4.45
3	Conj. Hab. Carlos Cueto Fernandini	88.74	84.96	−3.78	−4.26
4	Urb. El Trébol	73.13	129.60	56.47	77.22
5	Urb. Parque Naranjal	157.94	94.08	−63.86	−40.43
6	Urb. Prolima	90.79	83.95	−6.84	−7.53
7	Urb. San Elías	125.05	97.59	−27.46	−21.96
8	Urb. Santa Rosa de Lima (Virgen de la Puerta)	173.22	117.00	−56.22	−32.46
9	Urb. Villa del Norte	8.98	88.02	79.04	880.18
10	Urb. Villa Los Angeles–Sol de Oro	102.48	95.08	−7.40	−7.22
11	Urb. 12 de Agosto (Mariscal R. Castilla)	7.07	73.84	66.77	944.41
	Average	105.94	100.55	−5.39	−5.09

Note: K3 = temporary accumulation/storage points; Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex. Variation (m) was calculated as the hexagonal-model distance minus the current-scenario distance. Change (%) was calculated relative to the current-scenario distance. Positive values indicate an increase in distance, whereas negative values indicate a reduction. Extreme percentage increases in some sectors are due to very low baseline distances in the current scenario.

3.1.7 Estimated truck collection time

To estimate collection time in a Christaller hexagonal scenario with 59 temporary accumulation or storage points, the K2–K3 distance was first determined using geospatial processing in ArcGIS 10.8. This distance was then integrated with municipal collection time records from the 16 urban zones of Los Olivos. The baseline collection times reflect operational differences across urban sectors, with shorter times in consolidated residential areas and longer times in higher-density or more commercially active areas. The municipal collection-time records for the 16 urban sectors of Los Olivos show a bimodal distribution (Table 10): 7 min in seven sectors (43.75%) and 15 min in nine sectors (56.25%). Based on these sector-level values, the arithmetic average collection time is 11.5 min per point.

Based on the sector-level arithmetic mean, the average estimated collection time decreases from 11.50 min in the current 23-K3 scenario to 6.79 min in the optimized 59-K3 hexagonal scenario, corresponding to a reduction of approximately 40.96%. This result suggests that the redistribution of K3 points can shorten K2–K3 service distances and improve the operational efficiency of MSW collection. The largest reductions are observed in Urb. Covida, Urb. Mercurio, Urb. Parque Industrial Naranjal, Urb. San Elías, and Urb. Previ Naranjal, indicating that the hexagonal configuration is particularly effective in sectors with longer baseline collection times or less favorable initial accessibility. The optimized scenario may support shorter operational routes and potentially lower logistical costs in the analyzed urban area.

3.2 Territorial and Population Coverage

3.2.1 Territorial coverage

In Los Olivos district, with an official area of 18.25 km², the 23 existing K3 temporary accumulation/storage points were georeferenced in the field using GPS surveying. These data were processed in ArcGIS 10.8 to generate territorial coverage models for the current and optimized scenarios. Under the current 23-K3 scenario, the temporary accumulation/storage points cover approximately 5.82 km², corresponding to about 32% of the official district area. In contrast, the optimized 59-K3 hexagonal model covers approximately 14.10 km², corresponding to about 77% of the district area. This represents an increase of approximately 142% in territorial coverage (Table 11).

The largest relative increases are observed in sectors with very low baseline coverage, such as Urb. Parque Industrial Naranjal (1043.20%) and Urb. Mercurio (888.06%). These results suggest that the 59-K3 hexagonal

configuration improves spatial continuity and reduces coverage gaps across the district. The expanded coverage may also support more efficient collection-route planning and greater spatial equity in MSW service provision (Figure 7).

Table 10. Estimated collection time under the current and hexagonal scenarios

No.	Urban Sector	Collection Time (min)		Change (%)
		Current Scenario (23 K3 Points)	Hexagonal Model (59 K3 Points)	
1	AA.HH. Laura Caller Ibérico	15	11.93	−20.47
2	AA.HH. Villa Mercedes (Urb. 28 de Julio)	15	12.25	−18.33
3	Conj. Hab. Carlos Cueto Fernandini	7	5.79	−17.29
4	Urb. Covida	15	5.71	−61.93
5	Urb. El Trébol	15	11.35	−24.33
6	Urb. Mercurio	15	6.20	−58.67
7	Urb. Parque Industrial Naranjal	15	6.32	−57.87
8	Urb. Parque Naranjal	7	4.01	−42.71
9	Urb. Previ Naranjal	7	3.12	−55.43
10	Urb. Prolima	7	3.81	−45.57
11	Urb. San Elias	15	6.63	−55.80
12	Urb. Santa Rosa de Lima (Virgen de la Puerta)	15	10.03	−33.13
13	Urb. Villa del Norte	7	4.06	−42.00
14	Urb. Villa Los Angeles–Sol de Oro	7	3.76	−46.29
15	Urb. Villasol	7	3.16	−54.86
16	Urb. 12 de Agosto (Mariscal R. Castilla)	15	10.50	−30.00
	Average	11.50	6.79	−40.96

Note: K3 = temporary accumulation/storage points; Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex. Values represent estimated sector-level collection times. The average values are arithmetic means across the 16 urban sectors. Change (%) was calculated relative to the current 23-K3 scenario. Negative values indicate a reduction in collection time under the optimized 59-K3 hexagonal scenario.

Table 11. Territorial coverage under the current 23-K3 and optimized 59-K3 scenarios

No.	Urban Sector	Territorial Coverage (km ²)		Change (%)
		Current Scenario (23 K3 Points)	Hexagonal Model (59 K3 Points)	
1	Urb. Prolima	0.952694	2.254581	136.65
2	Urb. San Elias	0.640074	2.224083	247.47
3	Urb. Villa del Norte	0.351017	0.621973	77.19
4	Urb. Parque Naranjal	0.418946	1.054645	151.74
5	Urb. Villasol	0.188377	0.867814	360.68
6	AA.HH. Laura Caller Ibérico	0.430126	0.616284	43.28
7	Conj. Hab. Carlos Cueto Fernandini	0.734637	1.000568	36.20
8	Urb. 12 de Agosto (Mariscal R. Castilla)	0.329547	0.558971	69.62
9	Urb. El Trébol	0.681894	1.073392	57.41
10	Urb. Previ Naranjal	0.113274	0.569244	402.54
11	Urb. Villa Los Angeles–Sol de Oro	0.193798	0.512610	164.51
12	Urb. Covida	0.056230	0.547433	873.56
13	Urb. Santa Rosa de Lima (Virgen de la Puerta)	0.378869	0.641355	69.28
14	AA.HH. Villa Mercedes (Urb. 28 de Julio)	0.245901	0.364364	48.18
15	Urb. Parque Industrial Naranjal	0.084216	0.962755	1043.20
16	Urb. Mercurio	0.022852	0.225791	888.06
	Total	5.822452	14.095863	142.09

Note: K3 = temporary accumulation/storage points; Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex. Territorial coverage was calculated in ArcGIS 10.8 by intersecting the influence areas of K3 points with the 16 urban sectors. Change (%) was calculated relative to the current 23-K3 scenario and compares the optimized 59-K3 hexagonal scenario with the current scenario.

As a theoretical estimate, territorial coverage can also be approximated as:

$$\text{Coverage} = \frac{N_{K3} \times A_{\text{hex}}}{A_{\text{total}}} \times 100$$

Using $N_{K3} = 59$, $A_{\text{hex}} = 0.234 \text{ km}^2$, and $A_{\text{total}} = 18.25 \text{ km}^2$, the theoretical coverage is approximately 76%. This value is close to the GIS derived coverage of approximately 77%, calculated from the total coverage area reported in Table 11, with the small difference resulting from spatial intersection and boundary treatment in ArcGIS.

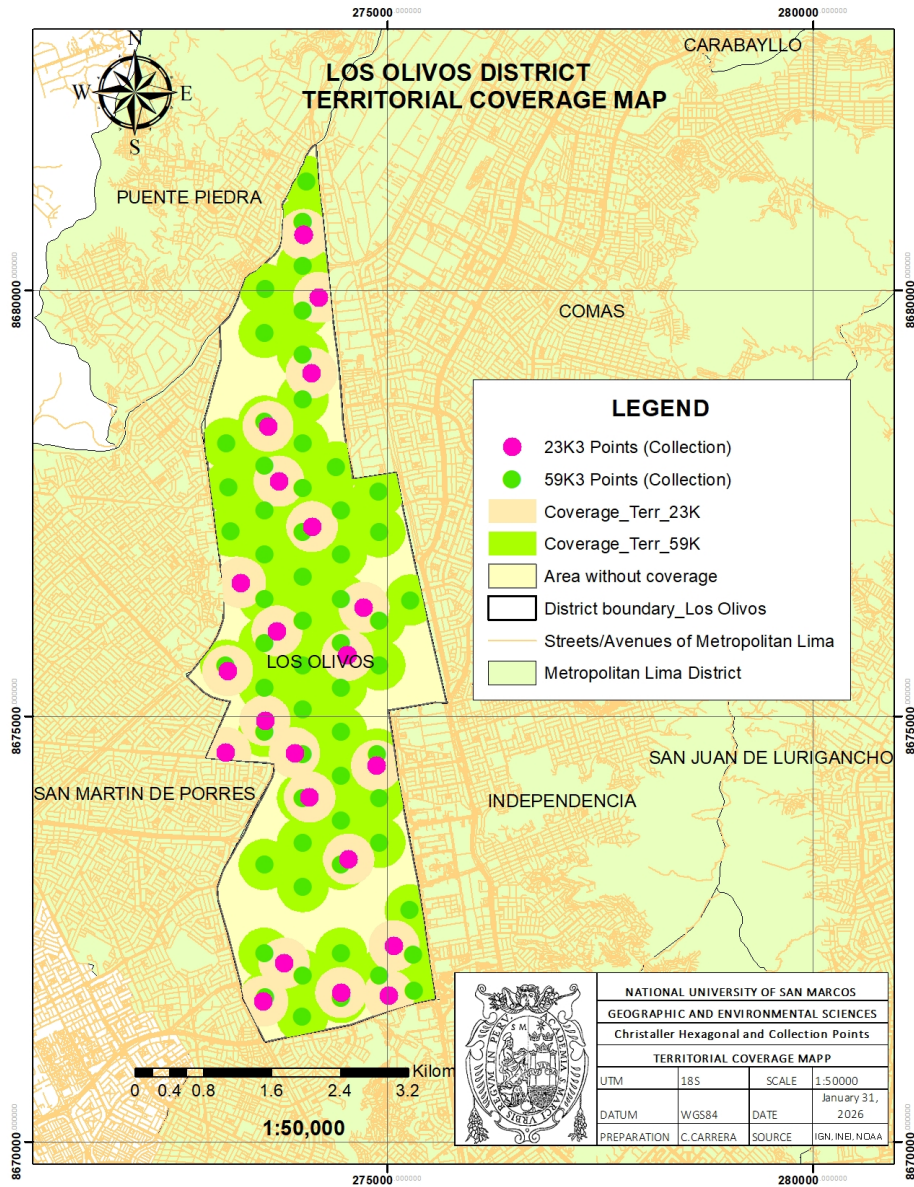


Figure 7. Territorial coverage under the current 23-K3 and optimized 59-K3 scenarios

Note: K3 = temporary accumulation/storage points.

3.2.2 Population coverage

Population coverage was estimated by spatially intersecting the service areas of the K3 points with the population distribution of the 16 urban sectors. The coverage percentage was calculated as:

$$\text{Population coverage (\%)} = \frac{P_{\text{covered}}}{P_{\text{total}}} \times 100$$

where, P_{covered} is the population located within the service area of the K3 points, and P_{total} is the total district population.

In ArcGIS 10.8, urban sector polygons were processed to estimate the population covered under the current 23-K3 scenario and the optimized 59-K3 hexagonal scenario. The area of each spatial unit was calculated using the Calculate Geometry tool and combined with the corresponding population density to estimate the number of inhabitants within each unit. The Buffer and Intersection tools were then applied to identify the population located within the K3 service areas. Through this spatial overlay analysis, the population coverage percentage for each scenario was determined, as reported in Table 12.

Based on Table 12, the optimized 59-K3 hexagonal model covers approximately 289,297 inhabitants, corresponding to 78.88% of the district population. In contrast, the current 23-K3 scenario covers approximately 121,660 inhabitants, corresponding to 33.17%. This indicates a substantial improvement in population coverage under the optimized scenario. The highest coverage levels under the 59-K3 model are observed in Urb. Villasol (97.10%), Urb. Previ Naranjal (95.70%), and Urb. San Elías (94.12%). In contrast, the current 23-K3 scenario shows very low coverage (<6%) in Urb. Mercurio, Urb. Parque Industrial Naranjal, and Urb. Covida. These differences suggest that the hexagonal model reduces population-coverage gaps and improves spatial equity in MSW service provision.

Table 12. Population coverage under the current 23-K3 and optimized 59-K3 scenarios

No.	Urban Sector	Total Population	Current Scenario (23 K3 Points)		Hexagonal Model (59 K3 Points)	
			Covered Population	Population Coverage(%)	Covered Population	Population Coverage (%)
1	Urb. Prolima	45,000	13,731.78	30.52	32,503.20	72.23
2	Urb. San Elías	43,850	11,873.07	27.08	41,273.05	94.12
3	Urb. Villa del Norte	29,135	14,659.91	50.32	25,982.57	89.18
4	Urb. Parque Naranjal	39,072	13,683.82	35.02	34,462.91	88.20
5	Urb. Villasol	22,029	4,640.88	21.07	21,389.92	97.10
6	AA.HH. Laura Caller Ibérico	18,640	10,485.19	56.25	15,026.68	80.62
7	Conj. Hab. Carlos Cueto Fernandini	16,741	9,516.91	56.85	12,964.02	77.44
8	Urb. 12 de Agosto (Mariscal R. Castilla)	18,217	7,517.61	41.27	12,753.58	70.01
9	Urb. El Trébol	26,280	13,031.58	49.59	20,518.11	78.08
10	Urb. Previ Naranjal	11,470	2,182.90	19.03	10,976.68	95.70
11	Urb. Villa Los Angeles–Sol de Oro	22,814	5,367.33	23.53	14,197.97	62.23
12	Urb. Covida	20,327	1,148.57	5.65	11,184.75	55.02
13	Urb. Santa Rosa de Lima (Virgen de la Puerta)	13,812	7,350.50	53.22	12,445.61	90.11
14	AA.HH. Villa Mercedes (Urb. 28 de Julio)	10,102	4,981.68	49.31	7,382.22	73.08
15	Urb. Parque Industrial Naranjal	19,237	980.38	5.10	11,211.83	58.28
16	Urb. Mercurio	10,025	508.23	5.07	5,024.39	50.12
	Total/overall coverage	366,751	121,660.34	33.17	289,297.49	78.88

Note: K3 = temporary accumulation/storage points; Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex.

Population coverage was calculated in ArcGIS 10.8 by intersecting K3 service areas with sector-level population data. The percentages represent the proportion of the population covered under each scenario. Total coverage was calculated as the total covered population divided by the total population, whereas average coverage refers to the arithmetic mean of sector-level coverage percentages.

3.3 Urban Impact

The urban impact was evaluated by estimating the average daily MSW load assigned to each K3 temporary accumulation/storage point. Before calculating this load, sector-level MSW generation was estimated using population, area, population density, and the adopted per-capita waste generation rate. First, the urban sector layer, population density layer, and waste-generation data were integrated in ArcGIS 10.8. The area of each sector was calculated using the Calculate Geometry tool. Population was then estimated by combining the calculated area with the corresponding population density, and the daily MSW generation was obtained by multiplying population by the per-capita waste generation rate. All calculations were performed as new fields in the attribute table and exported for comparative analysis (Table 13).

Population and area were used to estimate sector-level population density and MSW generation. The per-capita waste generation rate was applied to project daily waste volume. Total MSW quantifies daily collection needs per sector, enabling efficient resource allocation, vehicle deployment, and infrastructure sizing for sustainable waste management.

Urban impact was defined as the average daily solid waste load assigned to each K3 point, calculated as:

$$L_n = \frac{MSW_{total}}{n}$$

where, L_n is the average load per K3 temporary accumulation/storage point (tons/day), $MSW_{total} = 352.926$ tons/day is the total daily MSW generated in Los Olivos district, and n is the number of K3 temporary accumulation/storage points.

For the current 23-K3 scenario, the average load is:

$$L_{23} = \frac{352.926}{23} = 15.34 \text{ tons / day}$$

For the optimized 59-K3 hexagonal scenario, the average load is:

$$L_{59} = \frac{352.926}{59} = 5.98 \text{ tons / day}$$

The reduction in average load per K3 point was calculated as:

$$\text{Reduction} = \frac{15.34 - 5.98}{15.34} \times 100 = 61.03\%$$

Thus, the average daily load per K3 point decreases from 15.34 tons/day in the current 23-K3 scenario to 5.98 tons/day in the optimized 59-K3 scenario, corresponding to a reduction of approximately 61.03% (Table 14). This indicates that redistributing MSW across a larger number of K3 points can reduce the operational load assigned to each temporary accumulation/storage point.

Based on the quantitative indicators reported above, Table 15 summarizes the potential implications of the optimized 59-K3 hexagonal scenario.

Table 13. Generation of municipal solid waste (MSW) by urban sector

No.	Urban Sector	Population	Area (km ²)	Density (inhabitants/km ²)	Total MSW (tons/day)
1	Urb. Prolima	45,000	3.121	14,419	43.304
2	Urb. San Elias	43,850	2.363	18,558	42.197
3	Urb. Villa del Norte	29,135	0.697	41,777	28.037
4	Urb. Parque Naranjal	39,072	1.196	32,680	37.599
5	Urb. Villasol	22,029	0.894	24,649	21.199
6	AA.HH. Laura Caller Ibérico	18,640	0.764	24,385	17.937
7	Conj. Hab. Carlos Cueto Fernandini	16,741	1.292	12,958	16.110
8	Urb. 12 de Agosto (Mariscal R. Castilla)	18,217	0.798	22,820	17.530
9	Urb. El Trébol	26,280	1.375	19,118	25.289
10	Urb. Previ Naranjal	11,470	0.595	19,284	11.038
11	Urb. Villa Los Angeles–Sol de Oro	22,814	0.824	27,704	21.954
12	Urb. Covida	20,327	0.995	20,435	19.561
13	Urb. Santa Rosa de Lima (Virgen de la Puerta)	13,812	0.712	19,407	13.291
14	AA.HH. Villa Mercedes (Urb. 28 de Julio)	10,102	0.498	20,265	9.721
15	Urb. Parque Industrial Naranjal	19,237	1.651	11,648	18.512
16	Urb. Mercurio	10,025	0.450	22,258	9.647
	District overall	366,751	18.225	20,123.95	352.926

Note: Urb. = urbanization; AA.HH. = human settlements; Conj. Hab. = housing complex.

Table 14. Urban sectors of Los Olivos

Scenario	Average Load per Point K3 (tons/day)
Current model (23 K3 points)	15.34
Hexagonal model (59 K3 points)	5.98
Reduction	61.03%

Note: K3 = temporary accumulation/storage points.

Table 15. Potential implications of the optimized 59-K3 hexagonal scenario

Dimension	Potential Implication	Supporting Quantitative Evidence	Interpretation
Operational load at K3 points	Reduced operational pressure on temporary accumulation/storage points	Average load per K3 point: 15.34 → 5.98 tons/day, representing a reduction of 61.03%	A lower average load per point may reduce the risk of local saturation and improve operational manageability.
Territorial coverage	Expanded service coverage	Territorial coverage: 5.82 → 14.10 km ² , approximately 32% → 77%	The optimized layout reduces uncovered areas and improves spatial continuity.
Collection time	Shorter estimated collection time	Collection time: 11.50 → 6.79 min, representing a reduction of 40.96%	Shorter estimated collection time may support more efficient route planning.
K2–K3 accessibility	Reduced average service distance	K2–K3 distance: 413.78 → 224.38 m, representing a reduction of 45.76%	Shorter distances between K2 and K3 points may improve operational accessibility.
Population coverage	Increased population served	Covered population: 121,660 → 289,297 inhabitants, approximately 33.17% → 78.88%	Higher population coverage suggests improved service accessibility and spatial equity.
Urban environmental conditions	Potential reduction in localized waste accumulation	Lower load per K3 point and expanded territorial coverage	These changes may help reduce localized waste accumulation, although odors, vectors, and complaints were not directly measured.

Note: K2 = primary collection points or containers; K3 = temporary accumulation/storage points. The hexagonal model may improve MSW management by reducing the operational burden on K3 points, expanding territorial and population coverage, decreasing estimated collection time, and improving accessibility and spatial equity.

3.4 Optimal Location of Temporary Accumulation/Storage Points

The optimal location of K3 points in Los Olivos was evaluated using a Christaller-based hexagonal tessellation with a circumradius of $R = 300$ m. As described in the methodology, this configuration generated 59 proposed K3 points for the optimized scenario. The 300 m radius was adopted as an operational design parameter to balance spatial coverage, accessibility, and service continuity within the effective urban area of the district.

The expansion from the current 23 K3 points to the optimized 59 K3 points improves several spatial and operational indicators. Table 16 summarizes the main comparative results between the current and optimized scenarios.

Table 16. Summary of comparative indicators under the current and optimized scenarios

Urban Indicator	Current Scenario (23 K3 Points)	Optimized Hexagonal Scenario (59 K3 Points)	Unit
K2–K3 distance	413.80	224.38	m
K3–main road distance	105.94	100.55	m
Collection time	11.50	6.79	min
Territorial coverage	5.822	14.096	km ²
Covered population	121,660.34	289,297.49	inhabitants
Average load per K3 point	15.34	5.98	tons/day

Note: K2 = primary collection points or containers; K3 = temporary accumulation/storage points. The values summarize the main comparative indicators under the current 23-K3 scenario and the optimized 59-K3 hexagonal scenario.

The optimized 59-K3 hexagonal scenario reduces the average K2–K3 distance, slightly reduces the average K3–main road distance, increases territorial and population coverage, and lowers the average MSW load assigned to each K3 point. These results suggest that the Christaller-based spatial configuration may improve service accessibility, reduce coverage gaps, and distribute operational load more evenly across Los Olivos.

The proposed MSW system is organized as a five-level hierarchy, linking household-level waste generation, primary collection, temporary accumulation/storage, composting, and transfer functions. In this study, the hierarchy

includes 81,541 K1 points, 4,584 K2 points, 59 K3 points, 7 K4 points, and 2 K5 points. The K3, K4, and K5 layers were overlaid with the Christaller-based hexagonal grid in ArcGIS 10.8 to construct the adapted hierarchy of the proposed MSW management system (Figure 8). The final cartographic representation of this hierarchy is shown in Figure 9.

In Christaller's central place theory, the $K = 3$ marketing principle assumes that each higher level triples the service area and increases the service radius by approximately $\sqrt{3} \approx 1.732$ times. For the 59 K3 hexagons covering the effective urban area of 13.688 km², this theoretical hierarchy would imply approximately 177 K2 hexagons and 531 K1 hexagons at the lower levels, and approximately 20 K4 hexagons and 7 K5 hexagons at the higher levels. However, in the Los Olivos district, these theoretical proportions were adjusted to reflect urban complexity, population density, road accessibility, land-use constraints, and MSW generation patterns. Therefore, the hierarchy shown in Figure 9 should be interpreted as an adapted Christaller-based spatial structure rather than as a strict application of the classical $K = 3$ model.

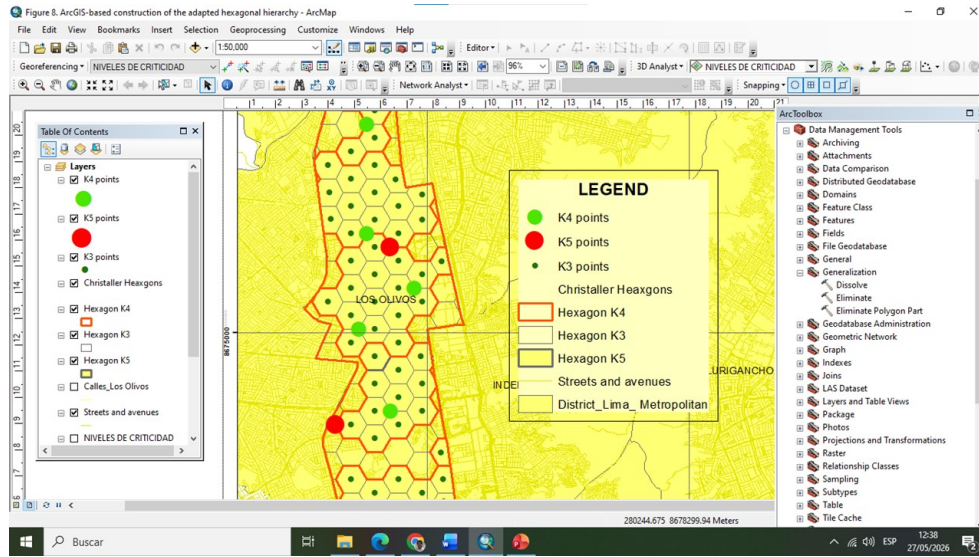


Figure 8. ArcGIS-based construction of the adapted hexagonal hierarchy

Note: The screenshot illustrates the ArcGIS 10.8 workflow used to construct the adapted K3–K5 hexagonal hierarchy.

Table 17. Statistical comparison of territorial and operational indicators between the current and optimized scenarios

Indicator	Scenario	Mean $\pm$ SD	Statistical Test	Statistic	df / n	p-Value	Interpretation
K2–K3 distance (m)	Current	413.78 $\pm$ 149.91	Paired Student's <i>t</i> -test	$t = 5.88$	$df = 15$	<0.001	Significant reduction (–45.8%)
Population coverage (%)	Current	33.06 $\pm$ 18.63	Wilcoxon signed-rank test	$W = 0$	$n = 16$	<0.001	Significant increase (+132.9%)
Collection time (min)	Current	11.50 $\pm$ 4.10	Wilcoxon signed-rank test	$W = 0$	$n = 16$	<0.001	Significant reduction (–40.96%)
Spatial correlation K2–K3	—	—	Pearson correlation	$r = 0.679$	$df = 14$	0.004	Moderate positive association; spatial pattern partially preserved

Note: K2 = primary collection points or containers; K3 = temporary accumulation/storage points; SD = standard deviation; *df* = degrees of freedom; *n* = number of paired sector-level observations. The population coverage values in this table are sector-level arithmetic means across the 16 urban sectors, rather than district-level overall coverage. “—” indicates not applicable. The Pearson correlation between sectoral K2–K3 distances under the current and optimized scenarios was recalculated as $r = 0.679$, $p = 0.004$, indicating a moderate positive association and suggesting that the optimized redistribution partially preserves the original spatial pattern while improving accessibility.

The statistical comparison between the current and optimized scenarios showed significant differences in the main territorial and operational indicators ($p < 0.01$). These results indicate that the optimized 59-K3 hexagonal scenario may improve territorial coverage, population coverage, K2–K3 accessibility, and estimated collection time

when compared with the current 23-K3 scenario. Furthermore, Pearson correlation analysis of K2–K3 distances at sector level showed a moderate-to-strong positive correlation between the two scenarios ($r = 0.679$, $p = 0.004$). This suggests that the optimized hexagonal configuration partially preserves the original spatial pattern while reducing average operational distances, indicating a coherent rather than random redistribution of collection points (Table 17). Furthermore, these statistical results support the effectiveness of the optimized hexagonal configuration in improving selected spatial and operational indicators, within the assumptions of the GIS-based model and the available municipal data.

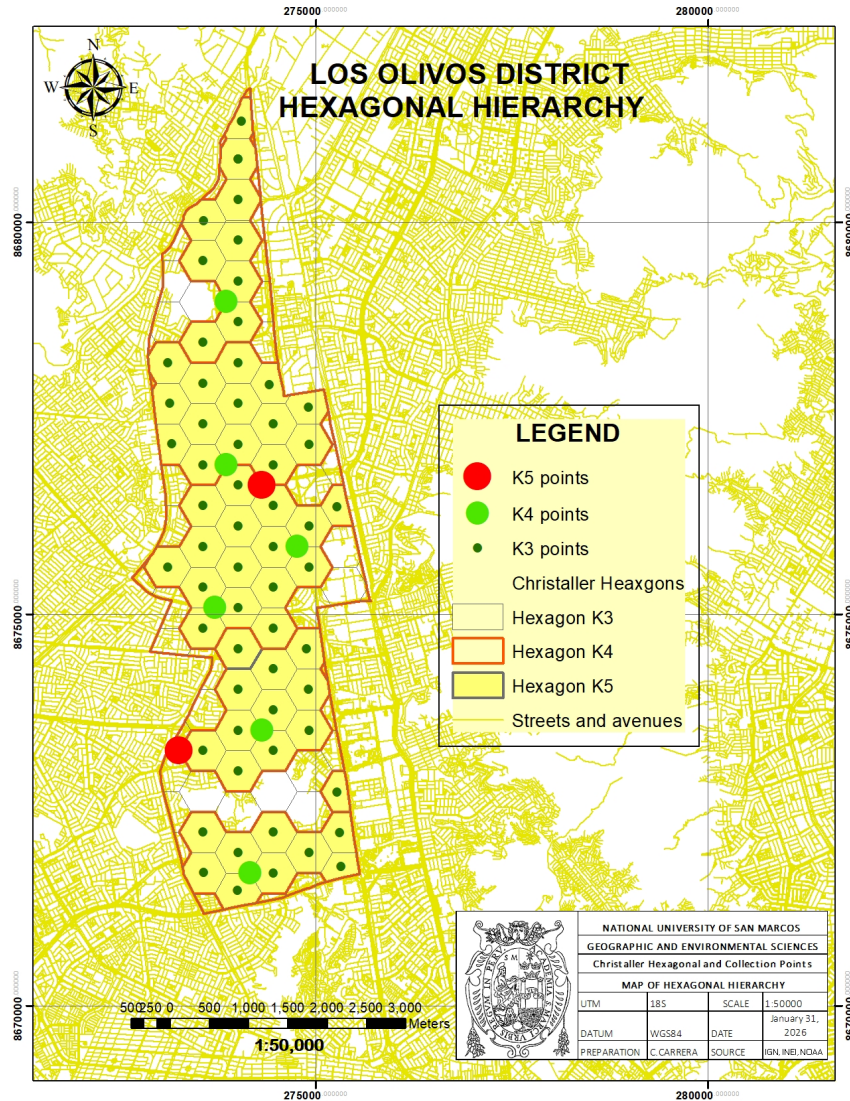


Figure 9. Adapted hexagonal hierarchy for Los Olivos

Note: K3 = temporary accumulation/storage points; K4 = composting points; K5 = transfer points. The map shows the adapted Christaller-based hierarchy of K3, K4, and K5 points in Los Olivos.

4 Discussion

4.1 Why the 59-Point Hexagonal Model Improves Coverage

The improvement in territorial coverage from 32% to 77% (Table 11) and population coverage from 33.1% to 78.88% (Table 12) stems from the geometric properties of the Christaller hexagon rather than merely increasing point density. A regular hexagonal tessellation with radius $R = 300$ m generates a service area of 0.234 km^2 per cell, supporting more continuous spatial coverage with reduced overlap and gaps compared with less regular spatial arrangements [9]. The theoretical calculation yielded 58.55 hexagons (rounded to 59), which aligns with the district's effective urban area of 13.688 km^2 . This geometric completeness explains why 5 sectors previously lacking K3 coverage under the current 23-point scenario now receive service. The reduction in average K2–K3 distance from 413.78 m to 224.38 m (45.76%) further indicates that hexagonal redistribution minimizes spatial friction by positioning

each point at the centroid of maximum accessibility within its Thiessen polygon, rather than at arbitrary locations.

4.2 Comparison with Previous Studies on Municipal Solid Waste Spatial Optimization

Our findings align with international research on facility location for waste collection. Vu et al. [21] analyzed parameter interrelationships in a dual-phase GIS-based MSW collection model, highlighting the importance of spatial and operational parameters in collection-system performance. Bautista and Pereira [7] demonstrated that evolutionary algorithms and Greedy Randomized Adaptive Search Procedure (GRASP) heuristics can optimize collection area location in Barcelona, reducing operational costs by 15–20%. Our Christaller-based approach achieves comparable efficiency gains (45.76% distance reduction, 61.03% reduction in average MSW load per K3 point) without requiring complex algorithmic computation, relying instead on deterministic geometric principles. Ghiani et al. [4] proposed a two-stage heuristic for recycling bin placement that minimized facility count while maintaining coverage; our model similarly balances coverage (77% territorial) with infrastructure investment (59 points). However, unlike these studies focused on facility minimization, our approach prioritizes territorial equity — ensuring no sector remains unserved. Herrera-Granda et al. [22] proposed a subregion districting model to optimize the MSW collection network in Ecuador. More directly, Shakya et al. [23] used GIS-based Network Analyst tools for MSW collection route optimization and reported a 12% reduction in collection time; by comparison, our micro-scale hexagonal optimization yields a 40.96% estimated reduction in sector-level collection time, suggesting that geometric pre-configuration of infrastructure may provide greater efficiency gains than post-hoc route optimization when the underlying spatial structure is heterogeneous.

4.3 Applicability and Limitations of the Christaller Model

The Christaller model provides a robust theoretical framework, but its application requires recognizing three key limitations. First, the model assumes an isotropic plain with a uniform population distribution, whereas Los Olivos exhibits marked density heterogeneity (11,648–41,777 inhabitants/km² in different sectors). We address this limitation by integrating a weighted overlay analysis that adjusted the hexagonal centroids toward areas of high criticality, thus adapting the theoretical grid to local spatial heterogeneity. Second, Christaller's original principle $K = 3$ (marketing) was designed for retail service areas, not for waste logistics. We adapted it by redefining K1–K5 as a waste generation-to-disposal hierarchy instead of market thresholds, maintaining the geometric logic but modifying the functional meaning. Third, the model assumes a continuous urban fabric, but Los Olivos contains non-urbanizable areas (mid-slopes, including a river, and industrial plots) that fragment the hexagonal grid. Our overlap with consolidated urban areas (75% of the district's total area) mitigates this problem by excluding non-urbanizable land. These adaptations suggest that the Christaller hexagon is applicable to environmental management when combined with GIS-based territorial filtering, as demonstrated by Lee and Lee [17] for the delineation of market areas in Seoul using GIS hydrological models.

4.4 Practical Implementation Constraints

Although the GIS simulation demonstrates a significant spatial improvement—increasing territorial coverage from 32% to 77% and reducing the operational distance by 45.76%—its actual implementation presents some limitations. The 59 hexagonal centroids represent mathematically optimal locations, but in the field, there may be restrictions due to private property, topography, or land availability. Furthermore, the model considers an average distance of 140 m to main roads using Euclidean criteria, without fully incorporating the actual conditions of urban traffic and road accessibility in Metropolitan Lima. Similarly, the simulation assumes a constant generation of 352.926 tons/day and a population of 366,751 inhabitants, despite the projected population growth for North Lima. Finally, the implementation of levels K4 and K5 would require inter-institutional coordination between municipalities and metropolitan environmental authorities.

4.5 Gap Between Geographic Information Systems Simulation and Real-World Implementation

Expanding the system from 23 to 59 collection points requires investment in containers, signage, and road improvements, as well as institutional coordination for its implementation. However, the 61% reduction in average operational load per K3 point suggests that the costs could be partially offset by potential reductions in collection time and route-related operating expenses in the medium term. Furthermore, there could be resistance from residents in residential areas and accessibility limitations in outlying sectors, despite the 300-m pedestrian radius considered in the model. Finally, expanding the system to the 43 districts of Metropolitan Lima would demand greater regulatory coordination and inter-municipal management capacity.

5 Conclusions

This study shows that the Christaller-based hexagonal model, integrated with GIS spatial analysis and demographic data, can support the optimization of MSW management in dense urban areas. In Los Olivos, the optimized 59-K3

scenario increased territorial coverage from approximately 32% to 77%, increased population coverage from 33.17% to 78.88%, reduced the average operational load per K3 point by 61.03%, and reduced the estimated sector-level collection time by 40.96%. The K1–K2–K3–K4–K5 hierarchy provides a spatial planning framework that may be adapted to other urban districts with similar MSW management challenges. However, practical implementation depends on land availability, economic feasibility, demographic dynamics, institutional coordination, and community acceptance. Future studies should incorporate dynamic simulations, economic evaluation, participatory GIS, and pilot-scale implementation to validate operational performance under real urban conditions.

Author Contributions

Conceptualization, B.G.H.F., C.E.C.S., C.F.C.C., and S.S.M.V.; methodology, B.G.H.F. and C.F.C.C.; software, C.E.C.S. and W.J.D.C.; validation, B.G.H.F., C.F.C.C., and C.E.C.S.; formal analysis, C.F.C.C. and C.E.C.S.; investigation, B.G.H.F., C.E.C.S., C.F.C.C., and S.S.M.V.; resources, B.G.H.F. and W.J.D.C.; data curation, C.E.C.S. and W.J.D.C.; writing—original draft preparation, B.G.H.F. and C.F.C.C.; writing—review and editing, S.S.M.V. and C.E.C.S.; visualization, C.E.C.S. and W.J.D.C.; supervision, C.E.C.S., B.G.H.F., and S.S.M.V.; project administration, B.G.H.F. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

MSW	Municipal solid waste
Urb.	Urbanization
AA.HH.	Human settlements
Conj. Hab.	Housing complex
MINAM	Ministry of the Environment (Ministerio del Ambiente del Perú)
INEI	National Institute of Statistics and Informatics (Instituto Nacional de Estadística e Informática)
OEFA	Environmental Assessment and Enforcement Agency (Organismo de Evaluación y Fiscalización Ambiental)
K1	Solid waste generation points
K2	Primary collection points or containers
K3	Temporary accumulation/storage points
K4	Composting points
K5	Transfer points